

- Q.26 What is electrode & explain Bio-electrodes.
 Q.27 Explain all types of electrodes.
 Q.28 Explain Resting & action potentials.
 Q.29 Explain about physiological systems of the body.
 Q.30 Explain Respiratory system of the human body.
 Q.31 Explain nervous system of the human body.
 Q.32 Write down the applications of computer in Biomedical system.
 Q.33 Write down about CAT scanners.
 Q.34 Write down the elements of ICU
 Q.35 Explain Spiro meter.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Explain any two instruction.
 i) Pacemaker
 ii) Bio-chemical system
 iii) Contact Impedence of electrode
 Q.37 Explain all about EEG & explain various factor to take care while taking this test.
 Q.38 Explain cardiovascular system & explain PQRS wave also.

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181561/121561

6th Sem / IC, EI Subject:- Bio-medical Instrumentation

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 The oxygenated blood flows in:-
 a) Veins b) Artries
 c) Both d) None
 Q.2 Longest bone of Human body:-
 a) Tibia b) Femur
 c) Radius d) Spine
 Q.3 A machine that records electrical activity of brain is known as
 a) EEG b) ECG
 c) EMG d) None
 Q.4 Upper chamber of Heart is called:-
 a) Atrium b) Ventricle
 c) Both d) None

Q.5 Full form of NIBP-

- a) Non Invasive Blood Pressure
- b) Non incentive Blood Pressure
- c) Not Invasive Blood Pressure
- d) Not Invasive Blood Pressure

Q.6 Spiro meter is used to measure :-

- a) Blood Pressure b) Respiratory Volume
- c) Blood Glucose d) Body Temperature

Q.7 Op-Amp is used for:-

- a) Signal conditioning b) For High CMRR
- c) High Gain d) All of these

Q.8 Signal to Noise Ratio should be :-

- a) Low b) High
- c) Both d) None

Q.9 Active transducers work on the principle of:-

- a) Energy Conversion b) Mass Conversion
- c) Energy Alteration d) Volume conversion

Q.10 Leucocyte are in the shape of-

- a) Sphere b) Cube
- c) Hollow d) Cuboid

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Normal Blood pressure of a Human body.

Q.12 What is the full form of ECG?

Q.13 What is Bio-electrode.

Q.14 Expand ICU.

Q.15 What is telemetry system.

Q.16 Two characteristics of OP-Amp

Q.17 Define Cell

Q.18 Write all four chamber of Heart.

Q.19 What is an amplifier.

Q.20 Describe about PQRS waveform of Heart.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 What is Man-instrumentation system and its components.

Q.22 What is Bio-telemetry & draw. the block diagram of Bio-telemetry system.

Q.23 Write a short note on ECG.

Q.24 Write a short note on Defibrillators.

Q.25 Write a short note on EMG.